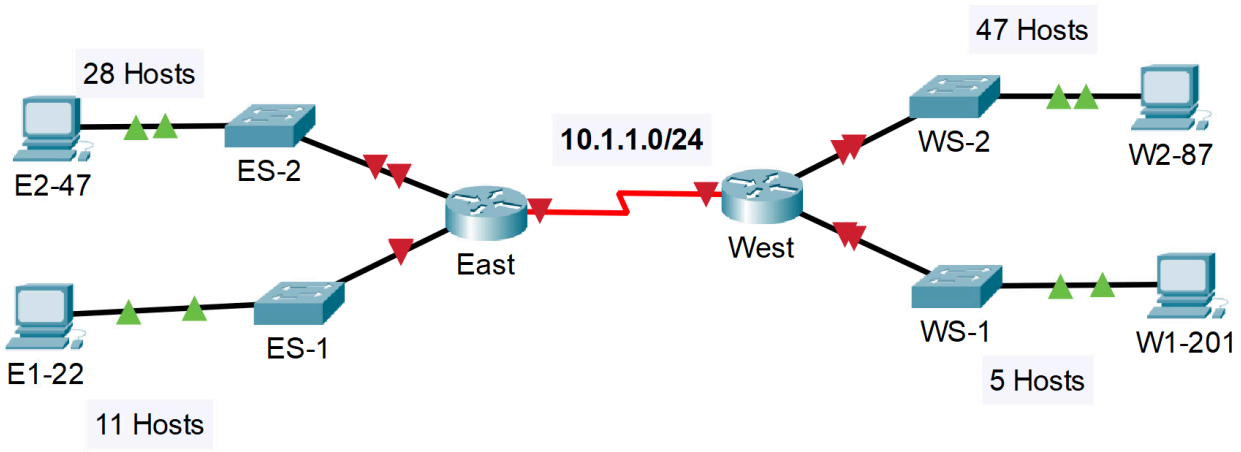
LAB 4 Part 1: Design and Implement VLSM Addressing



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
East Router	G0/0	10.1.1.65	255.255.255.224	N/A
	G0/1	10.1.1.97	255.255.255.240	N/A
	S0/0/0	10.1.1.121	255.255.255.252	N/A
West Router	G0/0	10.1.1.1	255.255.255.192	N/A
	G0/1	10.1.1.113	255.255.255.248	N/A
	S0/0/0	10.1.1.122	255.255.255.252	N/A
ES-1	VLAN 1	10.1.1.98	255.255.255.240	10.1.1.97
ES-2	VLAN 1	10.1.1.66	255.255.255.224	10.1.1.65
WS-1	VLAN 1	10.1.1.114	255.255.255.248	10.1.1.113
WS-2	VLAN 1	10.1.1.2	255.255.255.192	10.1.1.1
PC E1-22	NIC	10.1.1.110	255.255.255.240	10.1.1.97
PC E2-47	NIC	10.1.1.94	255.255.255.224	10.1.1.65
PC W1-201	NIC	10.1.1.118	255.255.255.248	10.1.1.113
PC W2-87	NIC	10.1.1.62	255.255.255.192	10.1.1.1

© 2019 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public

Objectives

In this lab you will design a VLSM addressing scheme given a network address and host requirements. You will configure addressing on routers, switches, and network hosts.

- Design a VLSM IP addressing scheme given requirements – Part 1
- Configure addressing on network devices and hosts – Part 2
- Verify IP connectivity – Part 2
- Troubleshoot connectivity issues as required – Part 2

Background / Scenario

You have been asked to design, implement, and test an addressing scheme for a customer. The customer has given you the network address that is suitable for the network, the topology, and the host requirements. You will implement and test your design.

Instructions

This scenario has two sites and a WAN link between them. Use the address space **10.1.1.0/24** to meet the requirements of the customer.

Pre-lab Guide

Variable-length subnet masking allows network engineers to organize hosts within a network so that traffic and security policies can be enforced at the network level.

What steps does an engineer follow to design the network of a new building?
An engineer first identifies the number of devices and hosts required in each LAN. Next, they find the available IP address range and calculate the appropriate subnet masks using VLSM.

Part 1: Design a VLSM Address Scheme

Host Requirements:

LAN	Number of Hosts
ES-1 LAN	11
ES-2 LAN	28
WS-1 LAN	5
WS-2 LAN	47

Design Requirements

- Create the addressing design. Follow guidelines provided in the curriculum regarding the order of the subnets.
- The subnets should be contiguous. There should be no unused address space between subnets.
- Provide the most efficient subnet possible for the point-to-point link between the routers.

© 2019 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public

www.netacad.com

- Document your design in the table below.

Subnet Description	Number of Hosts Needed	Network Address/CIDR	First Usable Host Address	Broadcast Address
ES-1 LAN	11	10.1.1.96/28	10.1.1.97	10.1.1.111
ES-2 LAN	28	10.1.1.64/27	10.1.1.65	10.1.1.95
WS-1 LAN	5	10.1.1.112/29	10.1.1.113	10.1.1.119
WS-2 LAN	47	10.1.1.0/26	10.1.1.1	10.1.1.63
WAN Link	2	10.1.1.120/30	10.1.1.121	10.1.1.123

Configuration Requirements

Note: You will configure addressing on **all** devices and hosts in the network.

- the serial WAN link.
- Assign the first usable IP addresses in the appropriate subnets to Site2's router for the two LANs links. Assign the remaining IP address for the WAN link to the Site2 router's serial interface.
- Assign the second usable IP addresses in the appropriate subnets to the switches.
- The switch management interface should be reachable from hosts on all of the LANs.
- Assign the last usable IP addresses in the appropriate subnets to the hosts.
- Configure each host with an available IP in the subnet, with the subnet mask of the subnet, and with the default gateway of the IP address of the router's interface within the subnet.

© 2019 - 2019 Cisco and/or its affiliates. All rights reserved. Cisco Public

www.netacad.com